

Basic Electrical Engineering

MODULE 2 — PART 1: COMPLETE NOTES

Module 2 (10 hrs): AC Circuits

For First-Year Students — written in simple, easy language

Course: Basic Electrical Engineering · Course Code: EE100

Module 2 — what it covers (as per the syllabus)

- Generation of sinusoidal voltage, frequency, average value, RMS value, form and peak factors
- Voltage and current relationship, with phasor diagrams, in R, L and C circuits; j operators
- Analysis of R-L, R-C and R-L-C series circuits; series and parallel resonance
- Real power, reactive power, apparent power and power factor; measurement of power
- Three-phase circuits: generation, balanced star (3-wire and 4-wire) and delta loads, phase-line relations, advantages

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1. Introduction to AC Circuits

1.1 Alternating Voltage

A voltage which **periodically changes its polarity** at regular intervals of time is called an **alternating voltage**. Similarly, an alternating current changes its direction periodically.

Why a sine waveform?

- The sine waveform produces the least disturbance in the electrical circuit and is the smoothest and most efficient waveform.
- A sinusoidal voltage applied to appropriately designed coils produces a revolving magnetic field which has the capacity to do work.
- The mathematical computations are much simpler with this waveform.
- By means of Fourier series analysis, any periodic function of whatever waveform can be represented in terms of sinusoids.

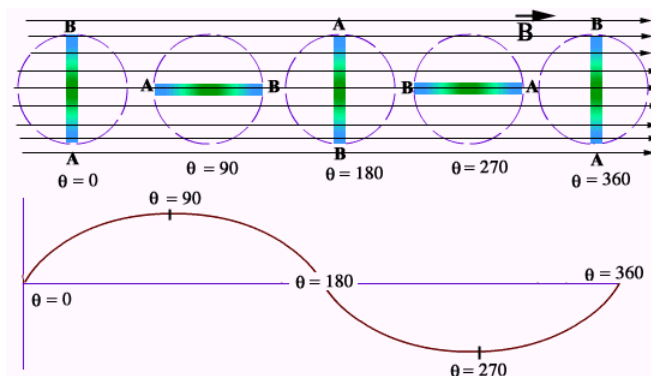


Figure 1: A coil rotating in a magnetic field generates a sinusoidal emf.

1.2 AC Terminology

Term	Meaning
Instantaneous value	The value of the alternating quantity at any instant — written as v or i .
Cycle	One complete set of positive and negative values of the alternating quantity.
Time period (T)	The time taken (in seconds) to complete one cycle.
Frequency (f)	The number of cycles per second, measured in hertz (Hz). One hertz = one cycle per second.
Amplitude (peak)	The maximum value (positive or negative) attained by the alternating quantity, V_m .

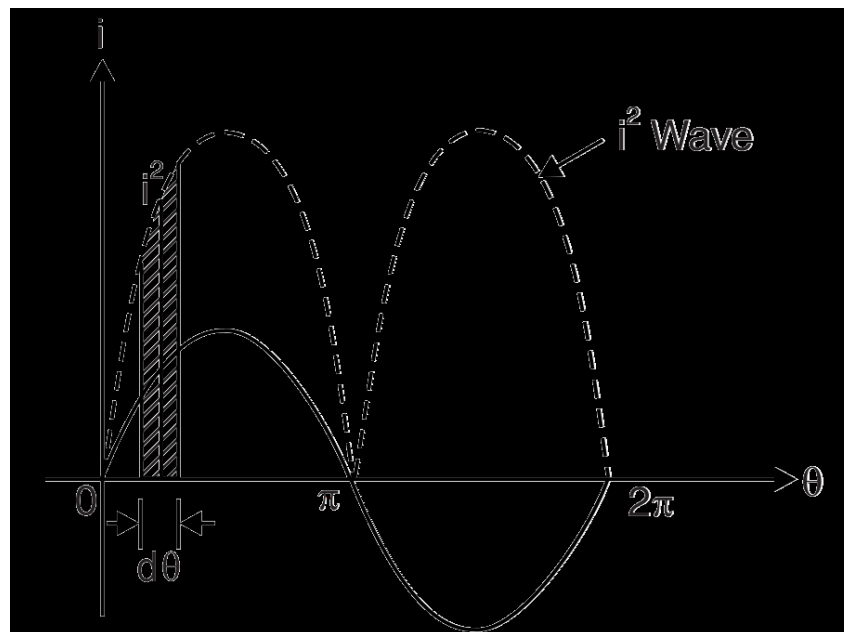


Figure 2: The sine waveform showing the cycle, peak value and time period.

Time period and frequency

$$f = 1 / T \quad T = 1 / f$$

Angular velocity

In one revolution the coil turns through 2π radians and the voltage wave completes one cycle. The time for one cycle is T , so the angular velocity is:

$$\omega = 2\pi / T = 2\pi f \quad (\text{radians per second})$$

1.3 Representation of an Alternating Quantity

In general, a sinusoidal voltage is written as:

$$v(t) = V_m \sin(\omega t + \phi)$$

- V_m is the amplitude of the sinusoid.
- ω is the angular velocity in radians per second.
- $(\omega t + \phi)$ is the argument and ϕ is the phase — both can be in radians or degrees.

Consider two sinusoids $v_1(t) = V_m \sin \omega t$ and $v_2(t) = V_m \sin(\omega t + \phi)$:

- The starting point of $v_2(t)$ occurs first in time, so $v_2(t)$ **leads** $v_1(t)$ by ϕ , or $v_1(t)$ **lags** $v_2(t)$ by ϕ .
- If $\phi \neq 0$ the two are **out of phase**; if $\phi = 0$ they are **in phase** (they reach their maxima and minima at the same time).
- They are compared in this way because they operate at the same frequency; they need not have the same amplitude.

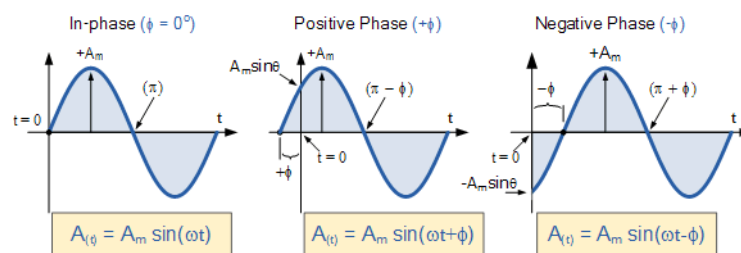


Figure 3: In-phase, leading and lagging sinusoids.

2. Values of an Alternating Quantity

There are four ways to express the magnitude of an alternating voltage or current: peak value, peak-to-peak value, average value and RMS value.

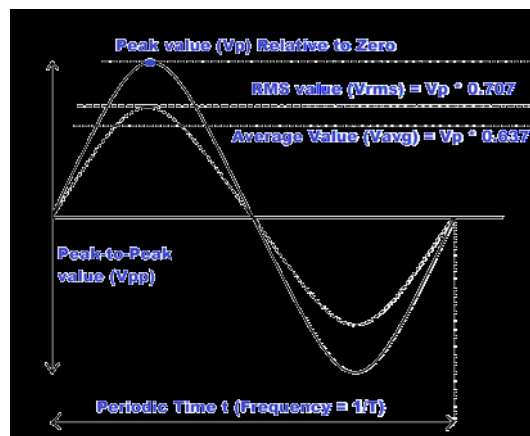


Figure 4: Peak, peak-to-peak, average and RMS values on the waveform.

2.1 Peak and Peak-to-Peak Values

The peak value is the maximum value V_m ; the peak-to-peak value is $2V_m$.

2.2 Average Value

The average value over one complete cycle is **zero**, because the waveform is symmetrical about the time axis and the positive area exactly cancels the negative area. The average value over a **half-cycle** is not zero, so "average value" means the half-cycle average unless stated otherwise. It is that steady (DC) value which would send the same charge in half a time period as the AC does.

$$\text{Average value } I_{av} = (\text{area of half-cycle})/(\text{base length}) = 2I_m/\pi = 0.637 I_m$$

2.3 RMS (Effective) Value

The RMS value of an alternating current is that steady (DC) current which, when flowing through a given resistance for a given time, produces the **same heat** as the alternating current flowing through the same resistance for the same time.

$$\text{RMS value } I_{rms} = I_m/\sqrt{2} = 0.707 I_m$$

2.4 Form Factor and Peak Factor

$$\text{Form factor} = (\text{RMS value})/(\text{Average value}) = (0.707 V_m)/(0.637 V_m) = 1.11$$

$$\text{Peak factor} = (\text{Maximum value})/(\text{RMS value}) = V_m/(0.707 V_m) = 1.414$$

Note: the domestic AC supply is 230 V, 50 Hz. This 230 V is the **RMS** value, so the peak value is $V_m = \sqrt{2} \times 230 = 325 \text{ V}$, and $v = 230\sqrt{2} \sin(314t)$ volts.

Quantity	Relation	Value
Peak-to-peak	$V_{p-p} = 2 V_m$	—
Average	$V_{av} = 0.637 V_m$	$2V_m/\pi$
RMS	$V_{rms} = 0.707 V_m$	$V_m/\sqrt{2}$
Peak from RMS	$V_m = 1.414 V_{rms}$	$\sqrt{2} V_{rms}$
Form factor	1.11	RMS/average

Quantity	Relation	Value
Peak factor	1.414	maximum/RMS

3. Concept of Phasors

Sinusoids are easily expressed in terms of **phasors**, which are more convenient to work with than sine and cosine functions. A sinusoidal alternating voltage or current is represented by a line of definite length rotating in the **anticlockwise** direction at a constant angular velocity ω . Such a rotating line is called a phasor. Its length equals the maximum value (on a suitable scale), and its angular velocity equals that of the alternating quantity.

3.1 Forms of Representing a Phasor

There are four ways of representing a phasor mathematically:

- **Rectangular form:** $V = a + jb$, where $j = \sqrt{-1}$, a is the real part and b the imaginary part; $V = \sqrt{a^2 + b^2}$ and $\theta = \tan^{-1}(b/a)$.
- **Trigonometrical form:** $V = V(\cos \theta + j \sin \theta)$, with $a = V \cos \theta$ and $b = V \sin \theta$.
- **Polar form:** $V = V \angle \theta$ (V is the magnitude, θ the phase angle measured anticlockwise from the reference axis).
- **Exponential form:** $V = V e^{j\theta}$, using Euler's equation $e^{j\theta} = \cos \theta + j \sin \theta$.

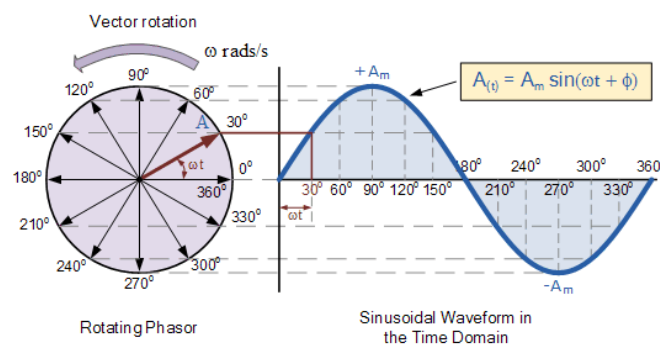


Figure 5: Rectangular components of a phasor.

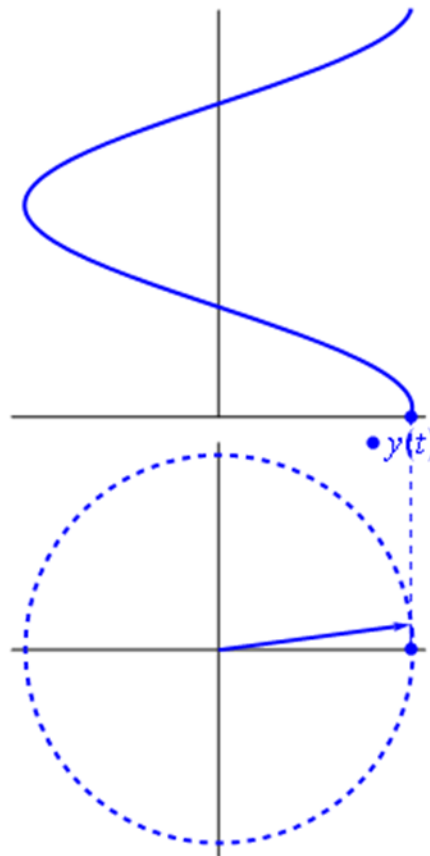


Figure 6: Polar and trigonometrical representation of a phasor.

3.2 Phasor Representation of a Sinusoid

Given a sinusoid $v(t) = V_m \cos(\omega t + \phi)$, it can be written as $v(t) = \text{Re}(V_m e^{j\phi} e^{j\omega t}) = \text{Re}(V e^{j\omega t})$, where $\mathbf{V} = V_m \angle \phi$ is the phasor representation. So a phasor is a complex representation of the magnitude and phase of a sinusoid:

$$v(t) = V_m \cos(\omega t + \phi) \Leftrightarrow \mathbf{V} = V_m \angle \phi$$

3.3 Basic Phasor Operations

Operation	Rule ($V_1 = V_1 \angle \theta_1$, $V_2 = V_2 \angle \theta_2$)
Addition	$V_1 + V_2 = (a_1 + a_2) + j(b_1 + b_2)$
Subtraction	$V_1 - V_2 = (a_1 - a_2) + j(b_1 - b_2)$
Multiplication	$V_1 V_2 = V_1 V_2 \angle (\theta_1 + \theta_2)$
Division	$V_1 / V_2 = (V_1 / V_2) \angle (\theta_1 - \theta_2)$
Reciprocal	$1/V = (1/V) \angle (-\theta)$
Square root	$\sqrt{V} = \sqrt{V} \angle (\theta/2)$

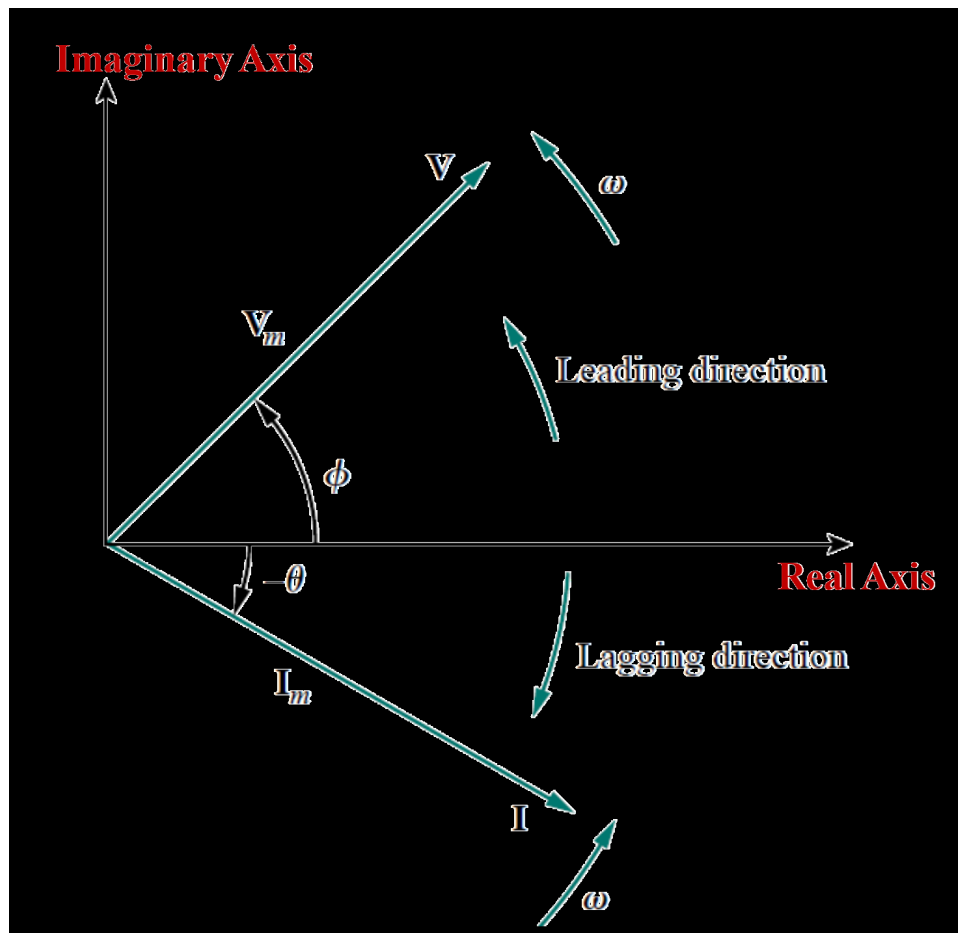


Figure 7: Phasor diagram and the four forms of representation.

4. Behaviour of R, L and C in an AC Circuit

4.1 Pure Resistance

With $v = V_m \sin \omega t$ across a pure resistance R , the current is $i = (V_m/R) \sin \omega t = I_m \sin \omega t$. So the current is **in phase** with the voltage. Impedance $Z = R$, and the power factor $= \cos 0^\circ = 1$.

4.2 Pure Inductance

In a pure inductor the current **lags** the voltage by 90° . The inductive reactance is $X_L = \omega L = 2\pi fL$, and the impedance is $Z = jX_L$. The power factor is zero, and the average power consumed is zero.

4.3 Pure Capacitance

In a pure capacitor the current **leads** the voltage by 90° . The capacitive reactance is $X_C = 1/(\omega C) = 1/(2\pi fC)$, and the impedance is $Z = -jX_C$. The power factor is zero, and the average power consumed is zero.

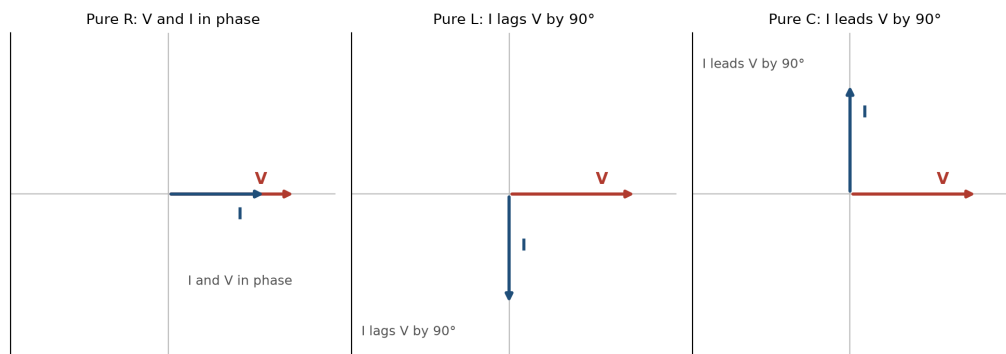


Figure 8: Phasor diagrams: R (in phase), L (I lags by 90°), C (I leads by 90°).

4.4 The j Operator

The letter j ($= \sqrt{-1}$) is called the **operator**: multiplying a phasor by j rotates it through $+90^\circ$ (anticlockwise), and multiplying by $-j$ rotates it through -90° . This is why:

$$Z_L = j\omega L \quad Z_C = 1/(j\omega C) = -j/(\omega C)$$

5. R-L, R-C and R-L-C Series Circuits

5.1 R-L Series Circuit

The same current flows through R and L. The voltage across R is in phase with the current, and the voltage across L leads the current by 90° . The applied voltage V is the phasor sum of V_R and V_L .

$$Z = R + jX_L \quad |Z| = \sqrt{R^2 + X_L^2} \quad \phi = \tan^{-1}(X_L/R)$$

The current lags the applied voltage by ϕ .

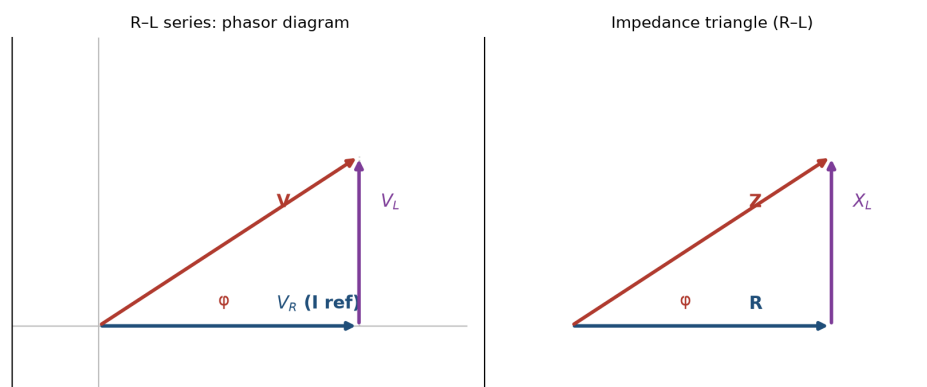


Figure 9: R-L series circuit: phasor diagram and impedance triangle.

5.2 R-C Series Circuit

The voltage across C lags the current by 90° . The current **leads** the applied voltage.

$$Z = R - jX_C \quad |Z| = \sqrt{R^2 + X_C^2} \quad \phi = \tan^{-1}(X_C/R)$$

5.3 R-L-C Series Circuit

The applied voltage is the phasor sum of V_R , V_L and V_C . The net reactance is $(X_L - X_C)$.

$$Z = R + j(X_L - X_C) \quad |Z| = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\phi = \tan^{-1}((X_L - X_C)/R) \quad \text{power factor} = \cos \phi = R/|Z|$$

- If $X_L > X_C$ the circuit is **inductive** and the current lags the voltage.
- If $X_C > X_L$ the circuit is **capacitive** and the current leads the voltage.
- If $X_L = X_C$ the circuit is **at resonance** and the current is in phase with the voltage.

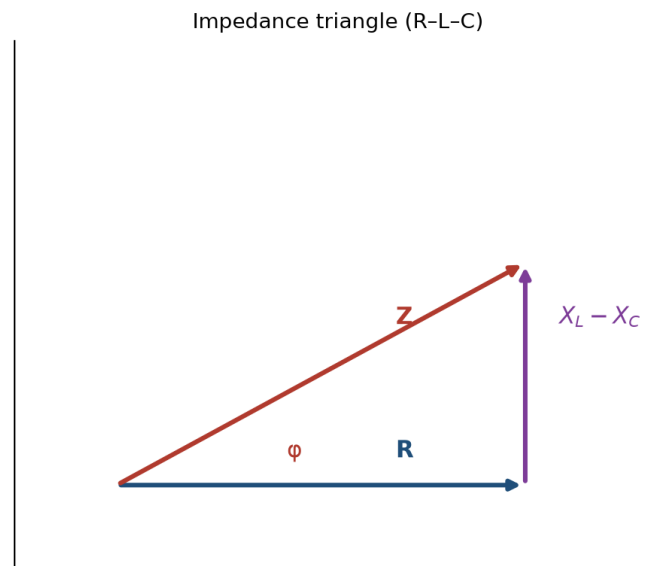


Figure 10: Impedance triangle of an R-L-C series circuit.

6. Resonance

6.1 Series Resonance

In a series R-L-C circuit, resonance occurs when the inductive reactance equals the capacitive reactance: $X_L = X_C$. At resonance:

- The impedance is minimum and purely resistive: $Z = R$.
- The current is maximum: $I = V/R$.
- The current and voltage are in phase, so the power factor is unity.
- The voltages across L and C can be much larger than the supply voltage (voltage magnification).

Resonant frequency: $f_0 = 1 / (2\pi \sqrt{LC})$

Quality factor: $Q = (1/R) \sqrt{L/C} = \omega_0 L / R$ Bandwidth = f_0 / Q

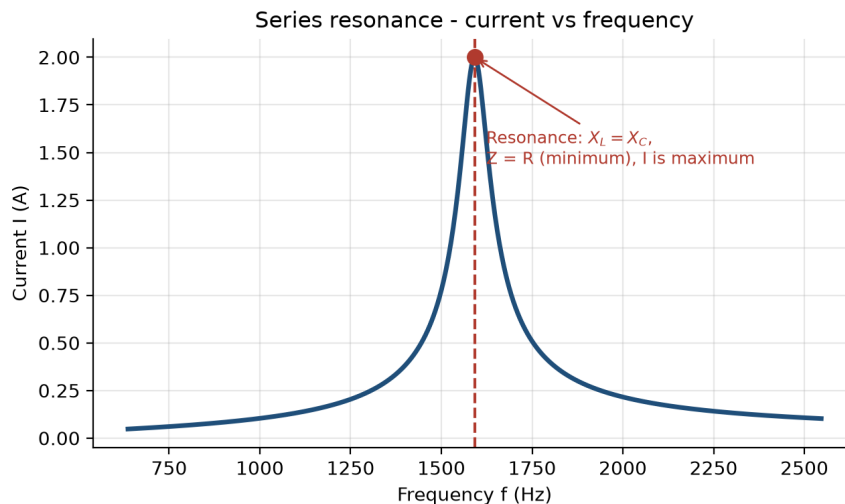


Figure 11: Series resonance — the current is maximum at the resonant frequency.

6.2 Parallel Resonance

In a parallel R-L-C circuit, resonance occurs when the reactive currents cancel. At resonance the impedance is **maximum** (ideally infinite) and the line current is minimum. For an ideal parallel circuit the resonant frequency is also $f_0 = 1/(2\pi\sqrt{LC})$.

7. Power in AC Circuits and Power Factor

In an AC circuit the instantaneous power is $p = vi$. Because of the phase difference ϕ between voltage and current, three kinds of power are defined:

Quantity	Formula	Unit	Meaning
Real (active) power P	$P = V I \cos \phi$	watt (W)	The power actually consumed
Reactive power Q	$Q = V I \sin \phi$	VAR	Power exchanged with L and C
Apparent power S	$S = V I$	volt-ampere (VA)	Product of RMS voltage and current

$$S = \sqrt{(P^2 + Q^2)} \quad \text{Power factor} = \cos \phi = P / S = R / |Z|$$

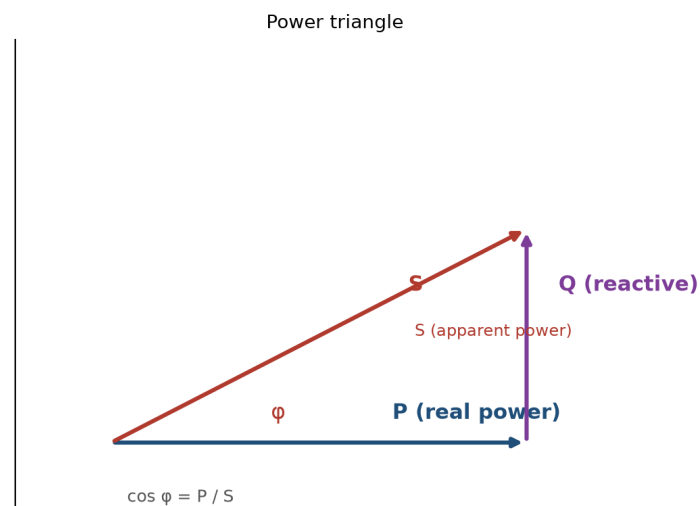


Figure 12: The power triangle relating P , Q and S .

- For a pure resistance: $P = VI$, $Q = 0$, $S = VI$, power factor = 1.
- For a pure inductance or capacitance: $P = 0$, $Q = VI$, power factor = 0 (no power is consumed).
- Power factor is **lagging** for inductive circuits and **leading** for capacitive circuits.

8. Measurement of Power

Power in an AC circuit is measured with a **wattmeter**, which is essentially a dynamometer-type instrument having two coils:

- **Current coil (CC)** — connected in **series** with the load; it carries the load current.
- **Potential (pressure) coil (PC)** — connected in **parallel** with the load through a high resistance; it carries a current proportional to the supply voltage.

The deflection of the wattmeter is proportional to $V I \cos \phi$, so it directly reads the real power. The reading is multiplied by the multiplying factor:

$$\text{Multiplying factor} = (\text{voltage range} \times \text{current range}) / \text{full-scale deflection}$$

9. Three-Phase Systems

9.1 Generation of Three-Phase Power

Circuits or systems in which the AC sources operate at the same frequency but different phases are called **polyphase** systems. In a **three-phase** system there are three equal voltages displaced from one another by 120° :

$$e_A = E_m \sin \omega t \quad e_B = E_m \sin(\omega t - 120^\circ) \quad e_C = E_m \sin(\omega t - 240^\circ)$$

The resultant $e_A + e_B + e_C = 0$ at every instant. The three phases are named after the colours Red (R), Yellow (Y) and Blue (B), giving the phase sequence RYB (there are only two possible sequences: RYB and RBY).

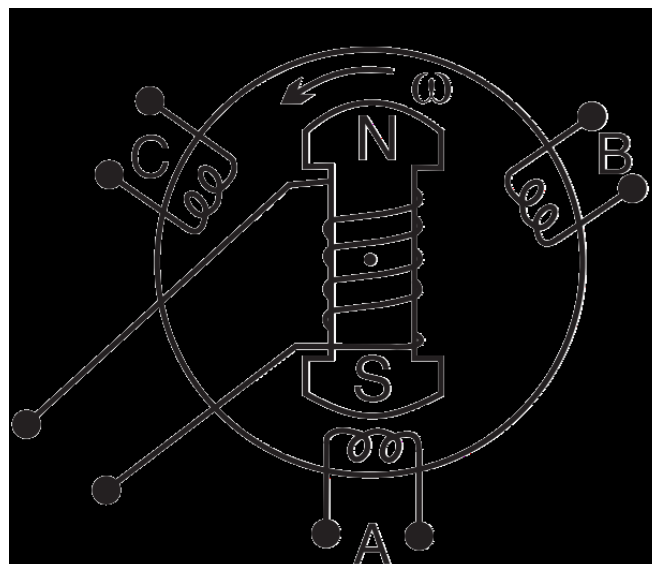


Figure 13: A three-phase generator with three coils 120° apart.

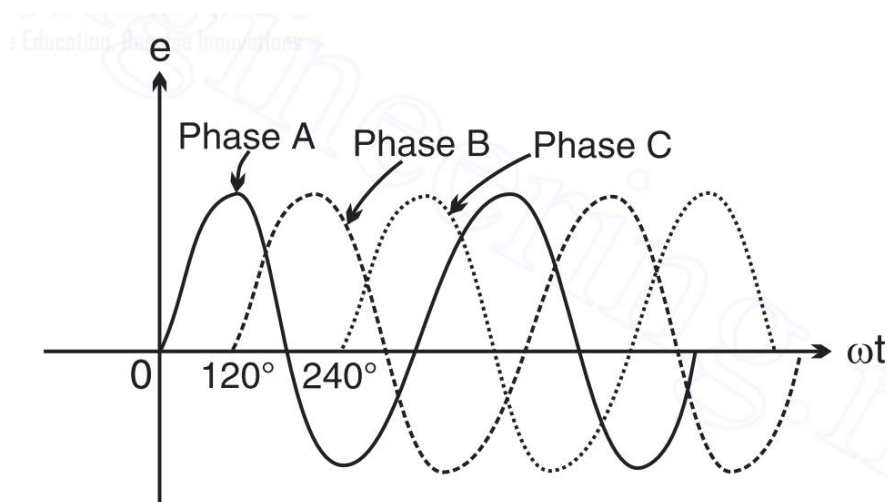


Figure 14: The three phase voltages displaced by 120° .

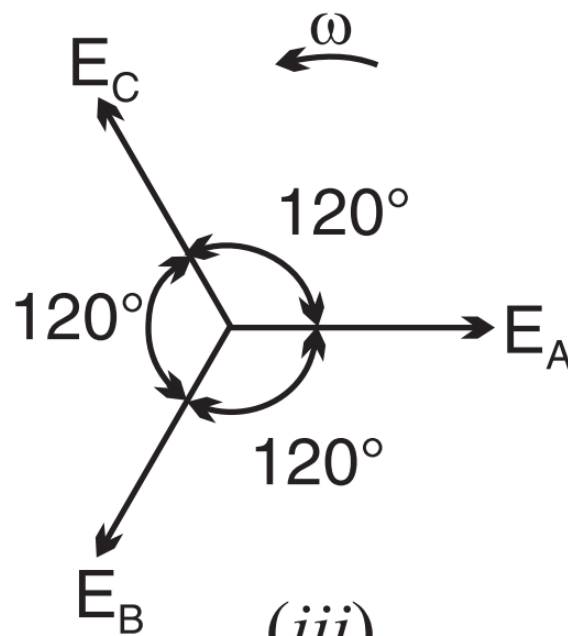


Figure 15: Phasor diagram of the three-phase voltages.

9.2 Star (Y) Connection

The three windings are joined at a common point called the **neutral**. If a neutral conductor is brought out, the system is a **3-phase, 4-wire** system; without the neutral it is a **3-phase, 3-wire** system.

$$V_L = \sqrt{3} V_{Ph} \quad I_L = I_{Ph}$$

Phase voltages are E_{RN} , E_{YN} , E_{BN} and line voltages are V_{RY} , V_{YB} , V_{BR} ; for a balanced supply $E_{RN} + E_{YN} + E_{BN} = 0$, and the line voltage leads the corresponding phase voltage by 30° .

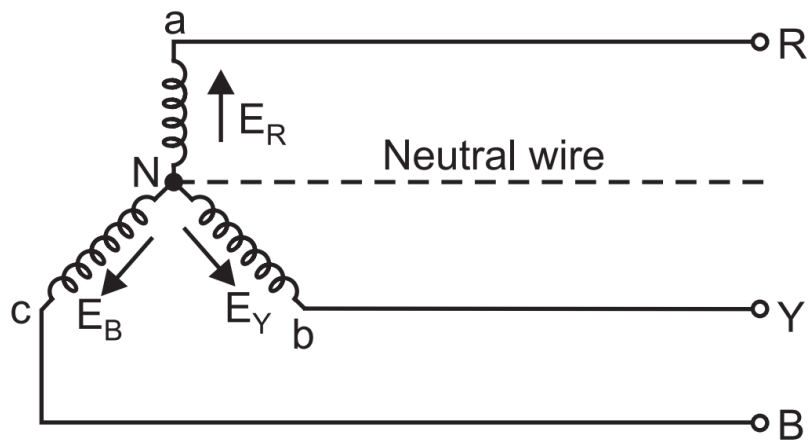
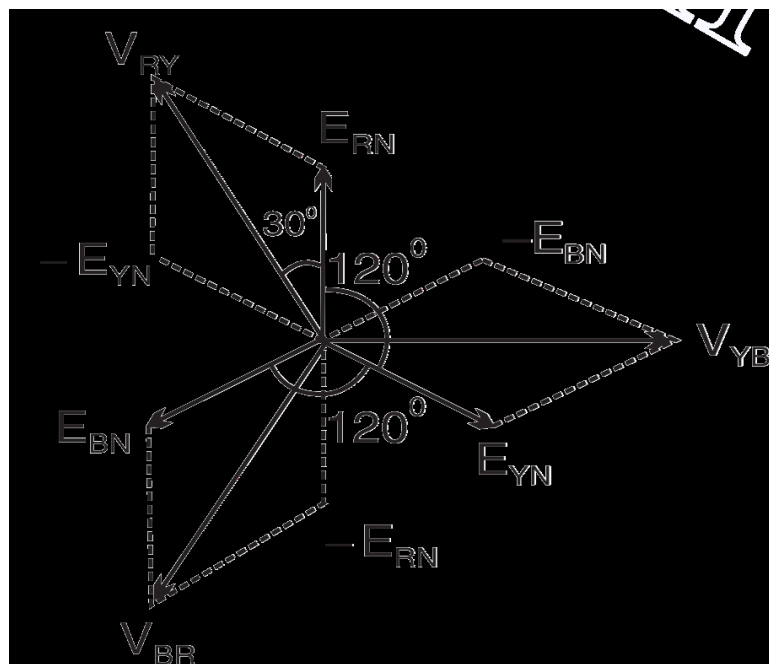


Figure 16: Star (Y) connected system.

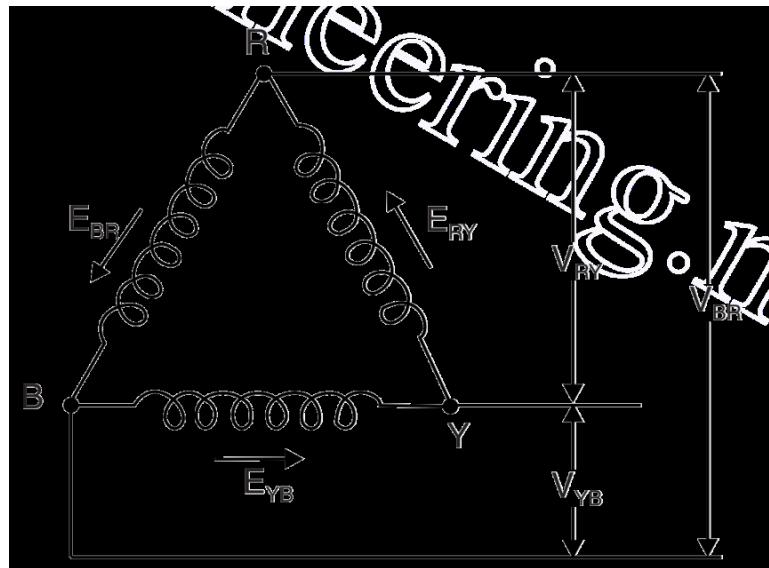
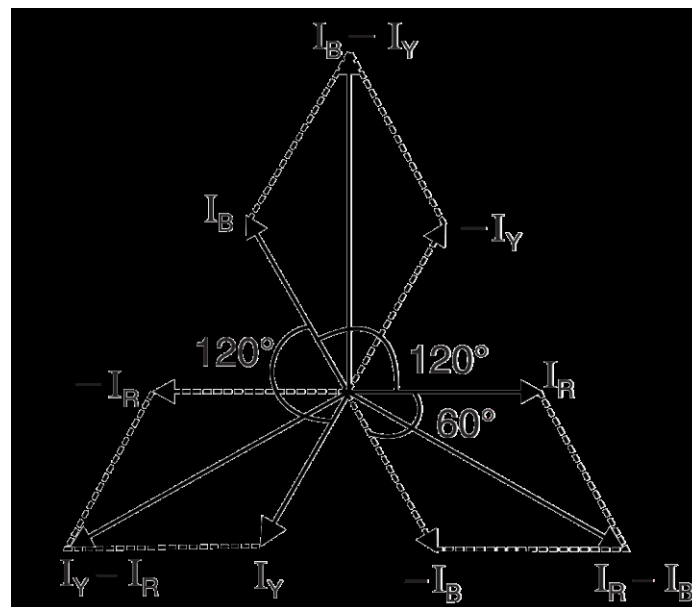
Figure 17: Phasor diagram showing $V_L = \sqrt{3} V_{ph}$.

9.3 Delta (Δ) Connection

The three windings are connected end-to-end in a closed loop. No neutral point exists, so only a 3-phase, 3-wire system can be formed.

$$V_L = V_{ph} \quad I_L = \sqrt{3} I_{ph}$$

The line current lags the corresponding phase current by 30° , and for a balanced system the line currents sum to zero.

Figure 18: Delta (Δ) connected system.Figure 19: Phasor diagram showing $I_L = \sqrt{3} I_{Ph}$.

9.4 Balanced Loads and Connections

A **balanced load** has equal phase impedances in magnitude and phase. For a balanced star-connected load $Z_1 = Z_2 = Z_3 = Z_Y$; for a balanced delta-connected load $Z_a = Z_b = Z_c = Z_\Delta$. The two are related by:

$$Z_\Delta = 3 Z_Y \quad Z_Y = Z_\Delta / 3$$

A three-phase source and load can each be wye- or delta-connected, giving four possible combinations: Y-Y, Δ -Y, Y- Δ and Δ - Δ .

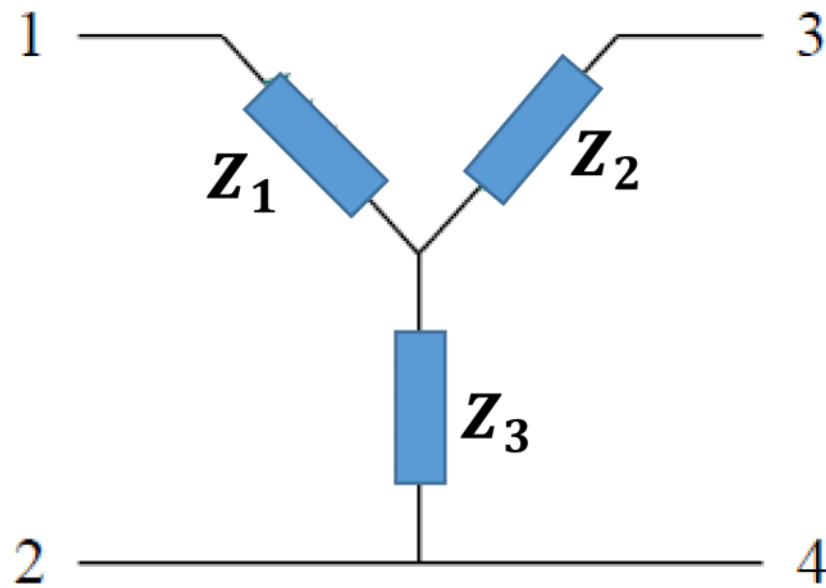


Figure 20: Balanced star-connected load.

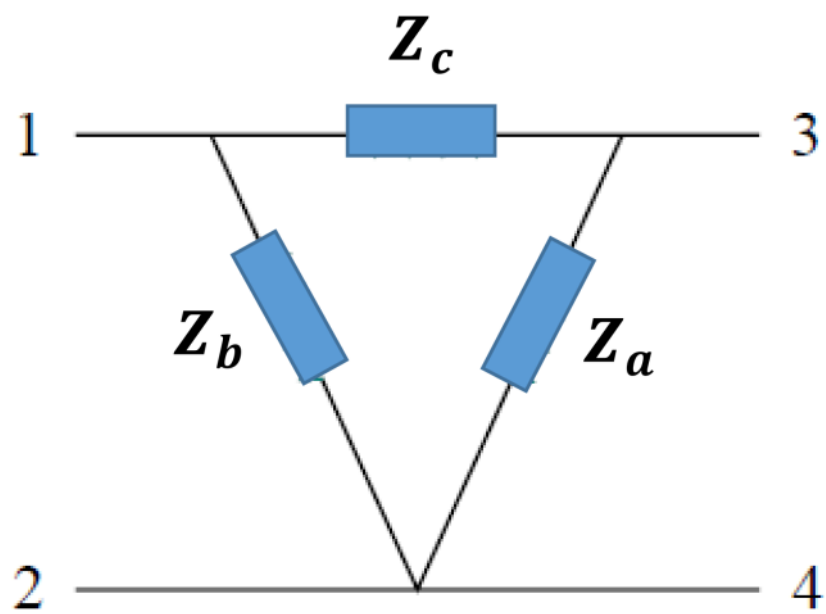


Figure 21: Balanced delta-connected load.

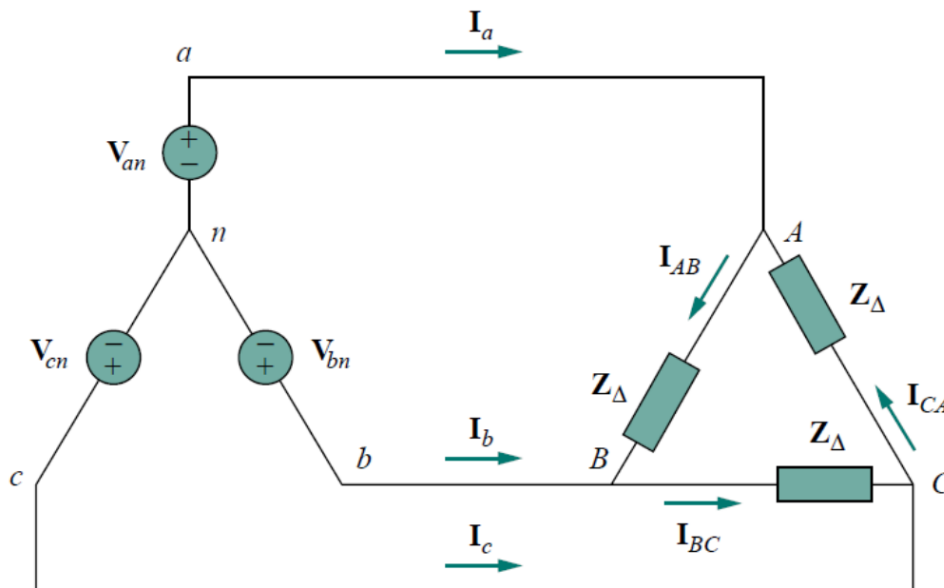


Figure 22: The four source-load connection combinations.

9.5 Three-Phase Power

Total power = 3 × power in each phase. In terms of line quantities (for both star and delta):

$$\text{Real power } P = \sqrt{3} V_L I_L \cos \phi$$

$$\text{Reactive power } Q = \sqrt{3} V_L I_L \sin \phi$$

$$\text{Apparent power } S = \sqrt{3} V_L I_L = \sqrt{(P^2 + Q^2)} \quad \text{Power factor} = \cos \phi = P/S$$

9.6 Why Three-Phase? Advantages

- **Constant power:** in a balanced 3-phase system the total power supplied at every instant is constant (in a single-phase circuit it varies sinusoidally).
- **Greater output:** a 3-phase machine gives more output than a single-phase machine of the same volume and weight.
- **Cheaper:** 3-phase motors are smaller and less expensive, using less copper, iron and insulation.
- **Transmission economics:** 3-phase power transmission is cheaper than single-phase, even though three conductors are needed instead of two.
- **Rotating magnetic field:** a 3-phase system sets up a rotating uniform magnetic field in stationary windings, which a single-phase current cannot do.

10. Solved Numerical Examples

Example 1: AC Values

Problem: find the amplitude, phase, period and frequency of the sinusoid $v(t) = 12 \sin(50t + 10^\circ)$.

$$\text{Amplitude } V_m = 12 \text{ V; phase } \phi = 10^\circ; \omega = 50 \text{ rad/s}$$

$$T = 2\pi/\omega = 2\pi/50 = 0.1257 \text{ s; } f = 1/T = 7.958 \text{ Hz}$$

Example 2: RMS and Average Values

Problem: a sinusoidal voltage has a peak value of 325 V. Find its RMS and average values.

$$V_{\text{rms}} = 0.707 \times 325 = 229.8 \text{ V} \approx 230 \text{ V}; \quad V_{\text{av}} = 0.637 \times 325 = 207 \text{ V}$$

Example 3: R-L Series Circuit

Problem: a coil of $R = 6 \, \Omega$ and $L = 0.05 \text{ H}$ is connected to a 230 V, 50 Hz supply. Find the impedance, current and power factor.

$$X_L = 2\pi fL = 2\pi \times 50 \times 0.05 = 15.7 \, \Omega$$

$$|Z| = \sqrt{(6^2 + 15.7^2)} = 16.8 \, \Omega; \quad I = 230/16.8 = 13.7 \text{ A}$$

$$\phi = \tan^{-1}(15.7/6) = 69.1^\circ \text{ (lagging)}; \quad \text{power factor} = \cos 69.1^\circ = 0.357$$

Example 4: Series Resonance

Problem: an R-L-C series circuit has $R = 10 \, \Omega$, $L = 10 \text{ mH}$ and $C = 1 \, \mu\text{F}$. Find the resonant frequency and the current at resonance for a 10 V supply.

$$f_0 = 1/(2\pi\sqrt{LC}) = 1/(2\pi\sqrt{(0.01 \times 10^{-6})}) = 1591.5 \text{ Hz}$$

$$\text{At resonance } Z = R = 10 \, \Omega, \text{ so } I = 10/10 = 1 \text{ A}$$

Example 5: Three-Phase Balanced Load

Problem: a Y-connected balanced three-phase generator with a phase voltage of 220 V supplies a balanced Y-connected load of $(10 + j5) \, \Omega$ per phase. Find the line voltage, phase current and total power.

$$V_L = \sqrt{3} \times 220 = 381 \text{ V}; \quad |Z| = \sqrt{(10^2 + 5^2)} = 11.18 \, \Omega$$

$$I_{\text{ph}} = 220/11.18 = 19.68 \text{ A} (= I_L \text{ for star})$$

$$P = 3 I^2 R = 3 \times 19.68^2 \times 10 = 11619 \text{ W}; \quad Q = 3 I^2 X = 5809 \text{ VAR}; \quad S = \sqrt{(P^2 + Q^2)} = 12990 \text{ VA}$$

11. Quick Reference — Key Formulae and Definitions

11.1 Key Formulae

Quantity	Formula	Notes
Frequency / period	$f = 1/T$; $\omega = 2\pi f$	ω in rad/s
Average value	$V_{av} = 0.637 V_m$	Half-cycle average
RMS value	$V_{rms} = 0.707 V_m = V_m/\sqrt{2}$	Effective value
Form factor	1.11	RMS/average
Peak factor	1.414	Maximum/RMS
Inductive reactance	$X_L = 2\pi fL$	Current lags by 90°
Capacitive reactance	$X_C = 1/(2\pi fC)$	Current leads by 90°
R-L-C impedance	$Z = \sqrt{R^2 + (X_L - X_C)^2}$	Series circuit
Series resonance	$f_0 = 1/(2\pi\sqrt{LC})$	$X_L = X_C$, $Z = R$
Power factor	$\cos \phi = R/ Z = P/S$	Lagging or leading
Real / reactive / apparent power	$P = VI \cos \phi$; $Q = VI \sin \phi$; $S = VI$	$S = \sqrt{P^2 + Q^2}$
Star connection	$V_L = \sqrt{3} V_{ph}$; $I_L = I_{ph}$	4-wire or 3-wire
Delta connection	$V_L = V_{ph}$; $I_L = \sqrt{3} I_{ph}$	3-wire only
Three-phase power	$P = \sqrt{3} V_L I_L \cos \phi$	$S = \sqrt{3} V_L I_L$

11.2 Key Definitions at a Glance

Term	Definition
Alternating quantity	One that periodically changes its polarity/direction at regular intervals.
Cycle / Time period	One complete set of positive and negative values / time for one cycle.
Frequency	Number of cycles per second (Hz).
Average value	Half-cycle average of the alternating quantity = $0.637 V_m$.
RMS value	The DC value producing the same heating effect = $0.707 V_m$.
Phasor	A rotating line whose length represents the maximum value and which rotates at ω .
j operator	$j = \sqrt{-1}$; multiplying by j rotates a phasor by $+90^\circ$.
Impedance	The total opposition to AC current, $Z = R + jX$ (in ohms).
Resonance	The condition $X_L = X_C$ at which current and voltage are in phase.
Power factor	$\cos \phi = P/S$; lagging for inductive, leading for capacitive circuits.
Wattmeter	An instrument that measures real power, using a current coil and a potential coil.
Balanced load	A load whose phase impedances are equal in magnitude and phase.
Phase sequence	The order in which the phase voltages reach their maxima (e.g. RYB).